

[作者] rogershan (Roger)

[看板] CSIE_exam

[標題] 計算機結構 期末考 鄭旭詠(101下)

[時間] Sun Jul 7 18:12:42 2013 to Sun Jul 7 18:31:19 2013

1. (20) Briefly and clearly explain Memory coherency and Memory consistency
2. (20) Suppose we would like to achieve a speedup of 10 with 40 processors.
What fraction of the original computation can be sequential?
3. (20) With the write back strategy, is it possible that we need to write the memory on a read miss? Explain why?
4. (20) Explain Compulsory Miss, Capacity Miss, and Conflict Miss in cache.
5. (20) The state transition diagrams of a write invalidate, cache coherence protocol for a write-back cache are illustrated below. Suppose two processors P1 and P2 are both in Shared state for a data block X.
P1 writes X at time T1. Then,
P2 reads X at time T2. Afterwards,
P2 writes X at time T3. Please write down and explain the state transitions and actions of P1 and P2 at time T1, T2, and T3.

圖1:

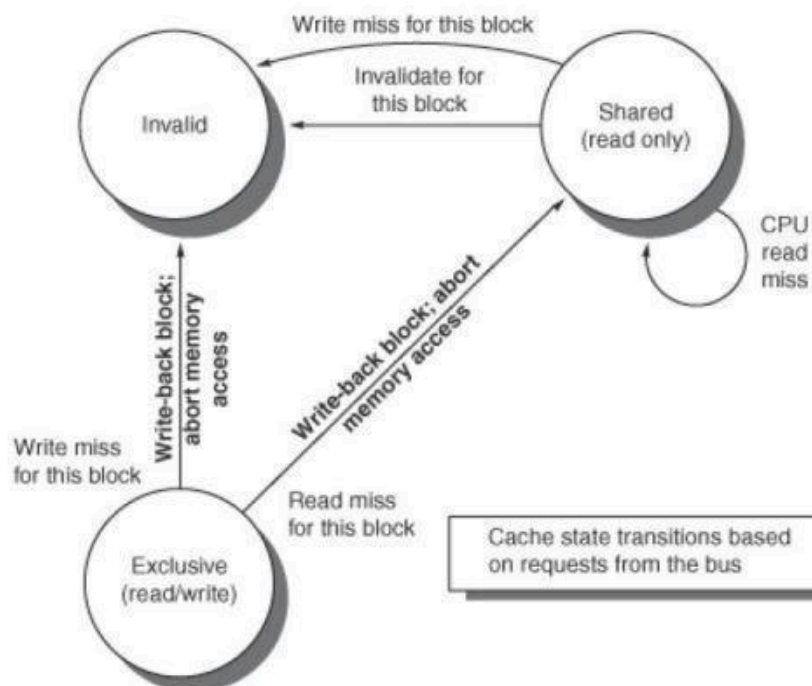
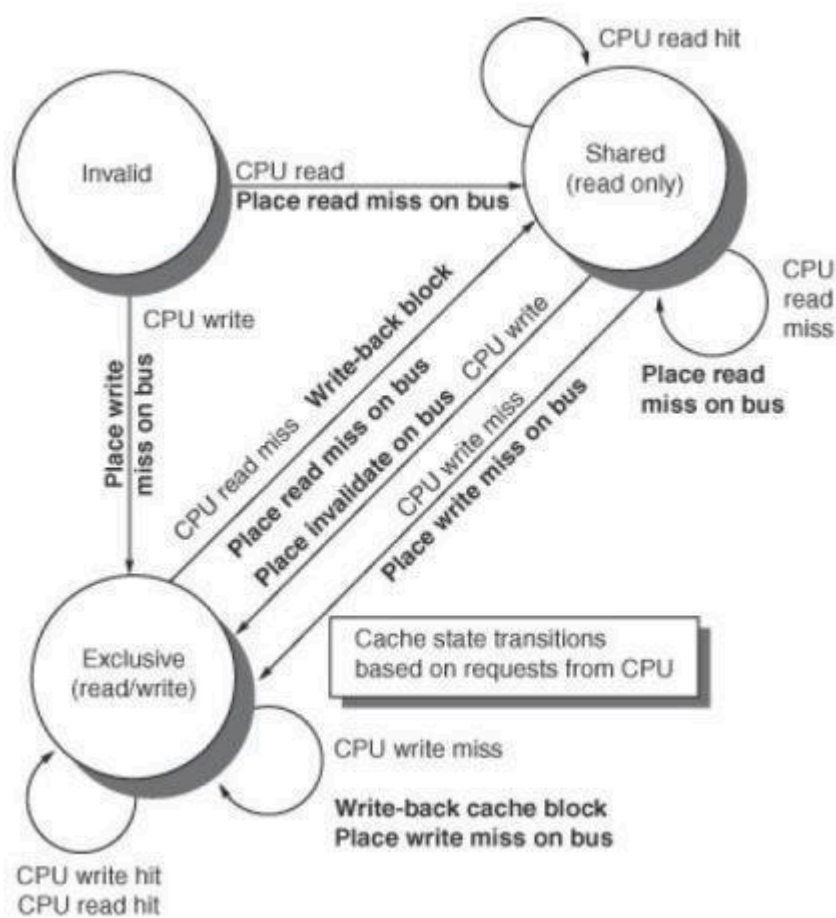


圖2:

ㄟ作者:rogershan 來自:220-137-110-229.dynamic.hinet.net
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